Overview

Purpose:
The in-game **Editor Overlay** and **RoomEditor** is built on ImGui and serves as the main level-editing tool for designers and programmers. The Imgui editor is now split into 2 editors, one for the main map (labryrinth) and another for the individual level rooms .

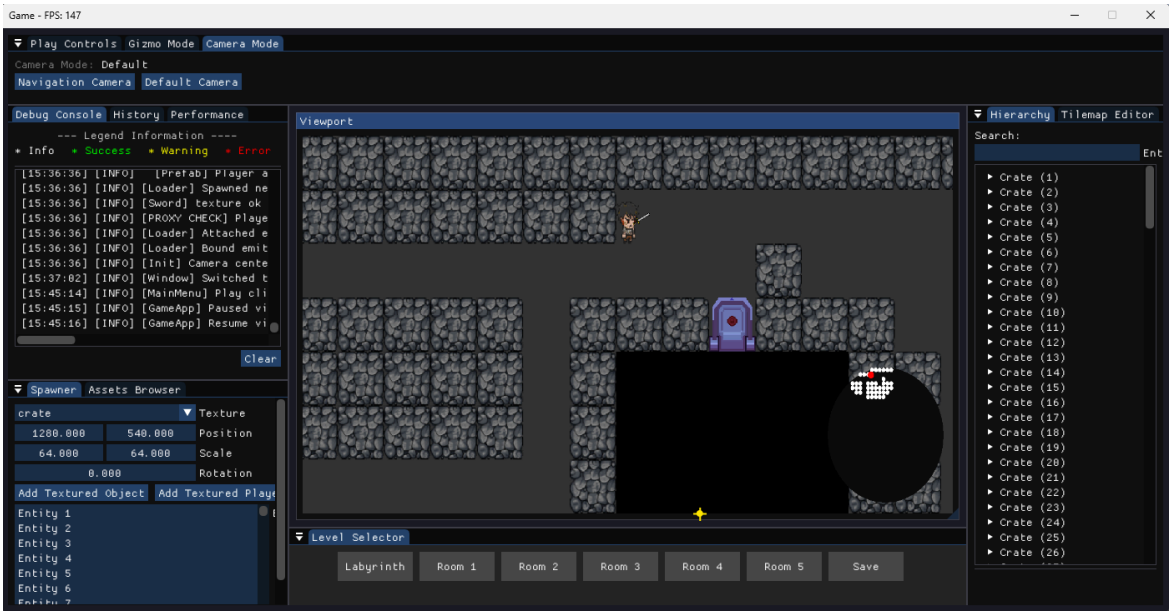
========== Editor Overlay ==========
==========
It allows users to:

- Inspect and modify existing entities
- Spawn and delete new entities
- Edit component data in real time
- Browse and apply textures
- Manage level transitions and saving
- Switch between *Editor* and *Play* modes
- Edit Tile map at runtime

The editor works on top of the ECS context, ensuring full access to all entities and components without depending directly on the GameApp runtime.

Interface Layout

Below is the layout of our Editor, toggled on and off via the f1 button, it is also auto docked

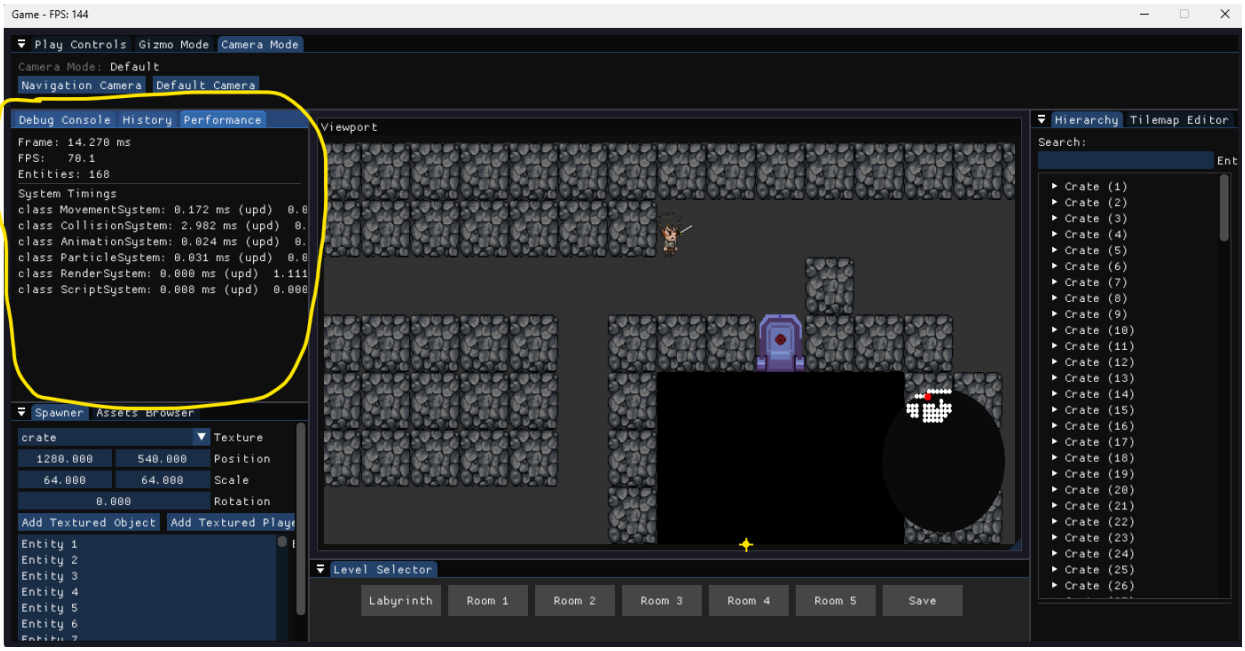


Panel Descriptions

1.0 Performance Panel

Top left corner.

Displays frame time, FPS, entity count, and per-system update timings.

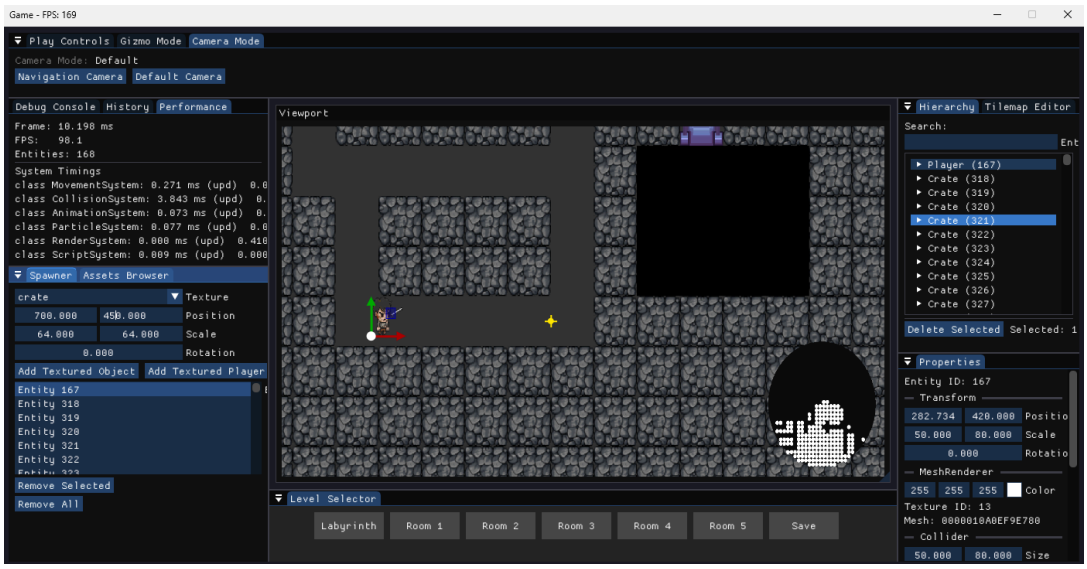


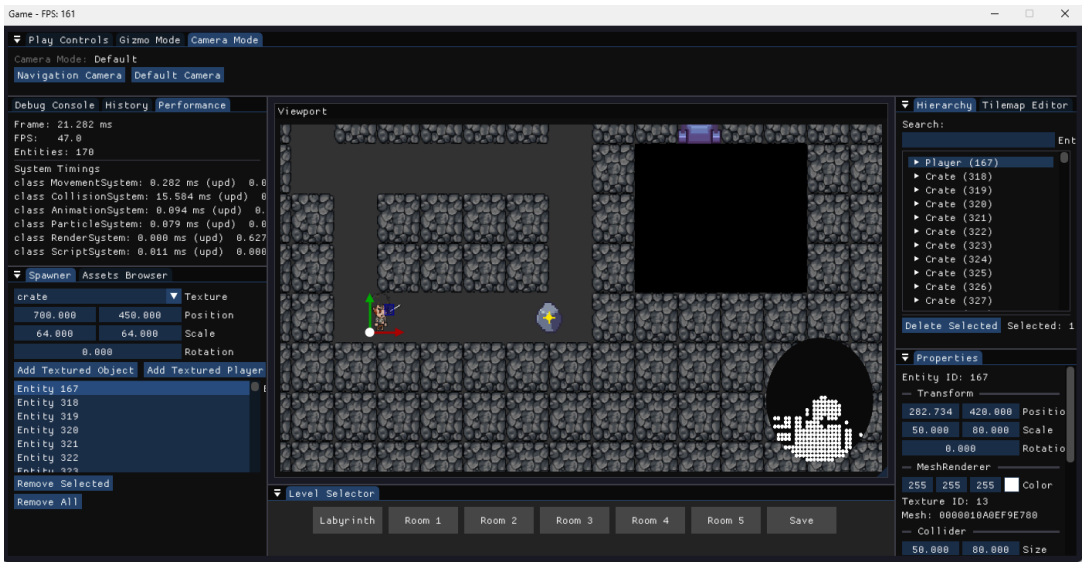
1.1 Spawner Panel

Located on the left side, this panel allows creating new entities. The spawn point is shown on screen via the yellow crosshair.

Usage:

1. Choose a texture type (e.g., `"crate"` or `"decorative"`).
2. Adjust position, scale, and rotation.
3. Click Add Textured Object to spawn into the scene.
4. Click Add Textured Player to spawn a controllable player prefab.





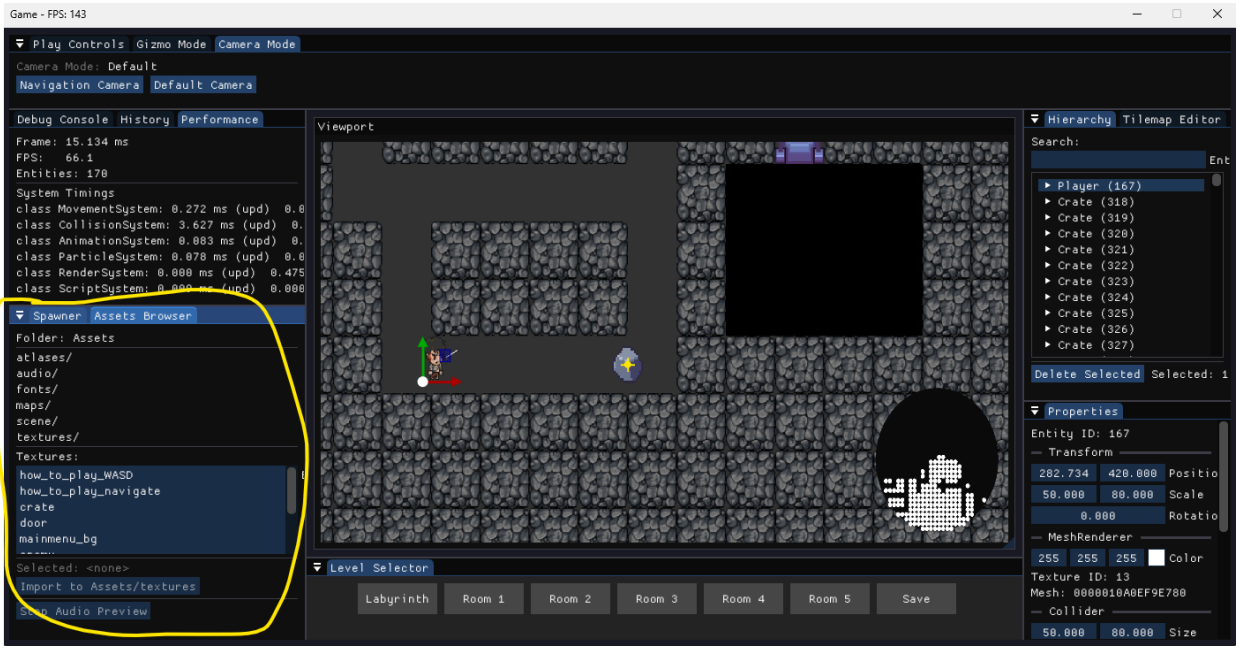
1.2 Assets Browser

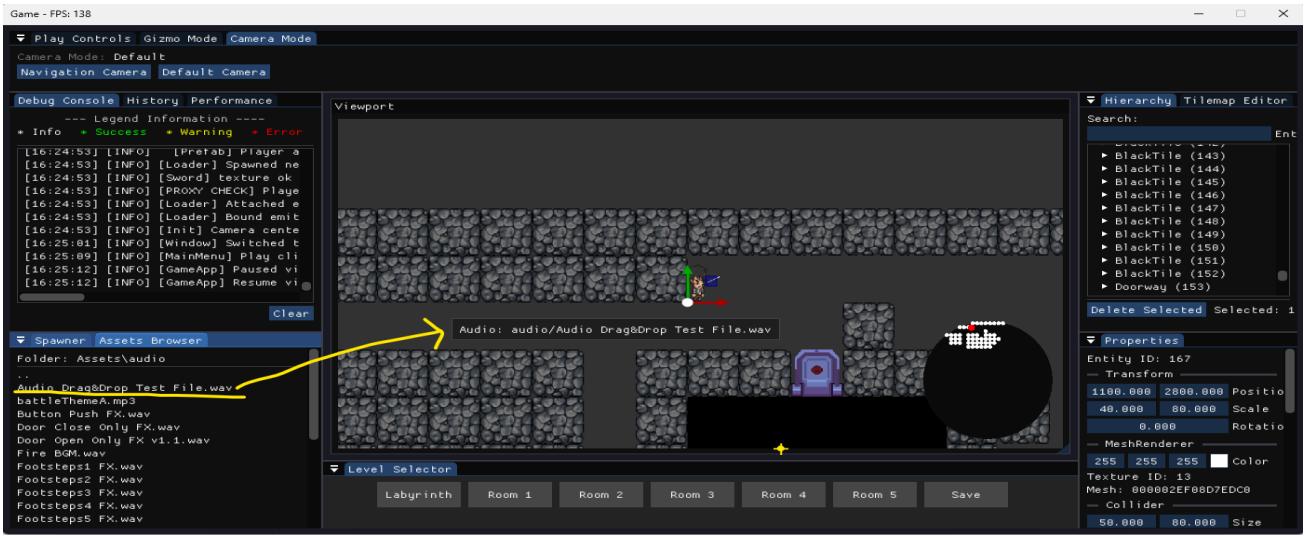
Located below the Spawner panel.

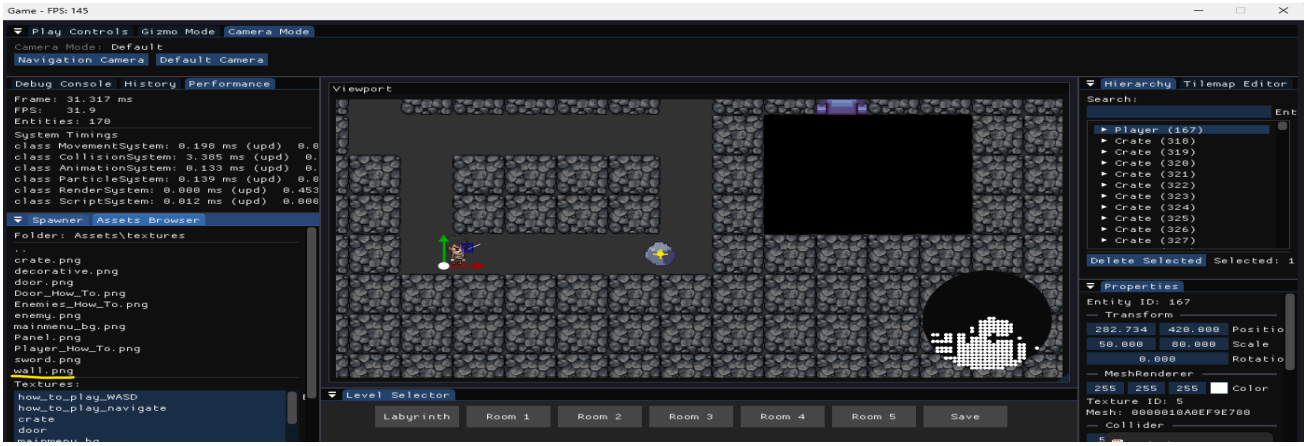
Purpose:
To import and manage textures used by entities.

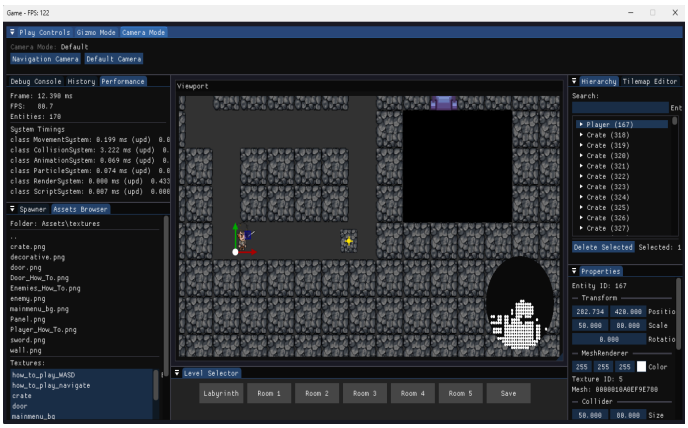
Usage:

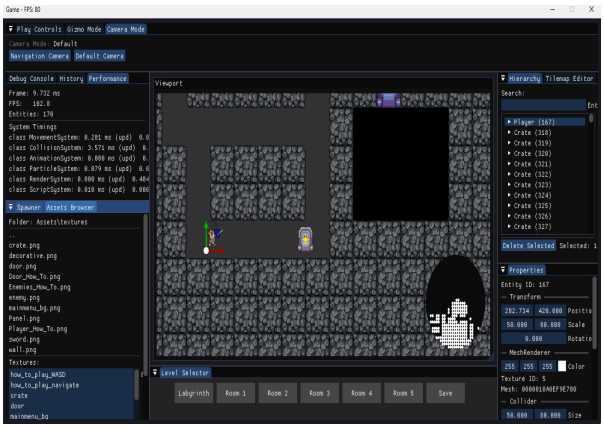
1. Browse folders under the Assets/ directory.
2. Select a .png file.
3. Click Import to Assets/textures.
4. The new texture appears under the Textures list.
5. Drag a texture from the list and drop it over any entity (except walls) in the viewport to apply it.
6. Drag an audio and drop it in the game viewport will allow the audio to run, press"stop audio preview" to stop











By dragging the newly added assets onto the texture, we can change the texture of our newly created wall

1.3 Hierarchy Panel

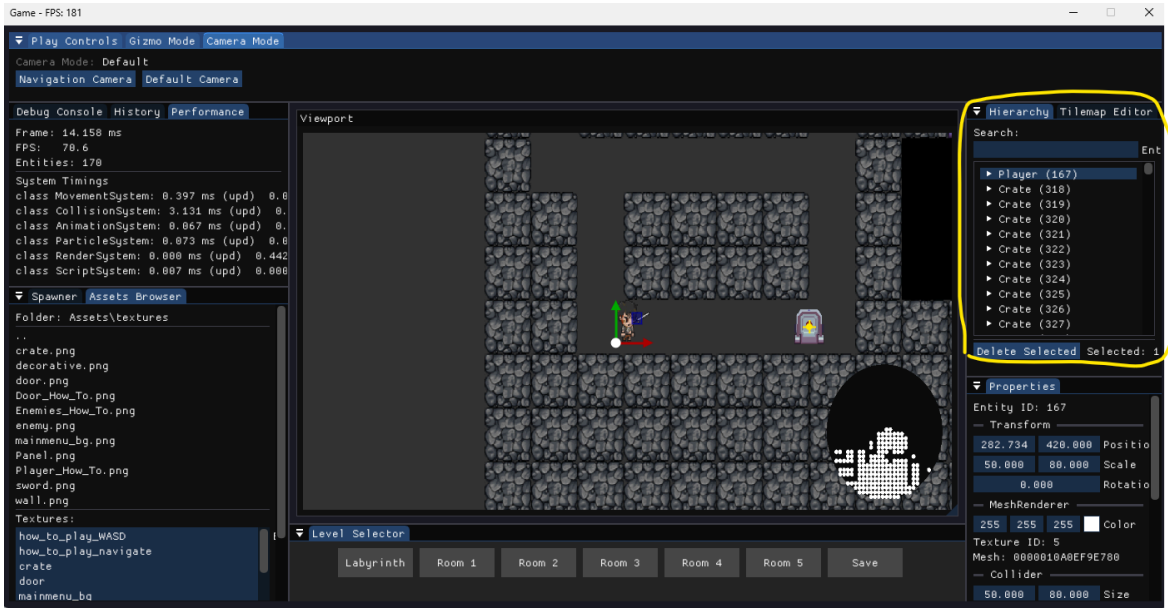
Right side of the editor.

Purpose:
Displays all active entities with a search bar and expandable component list.

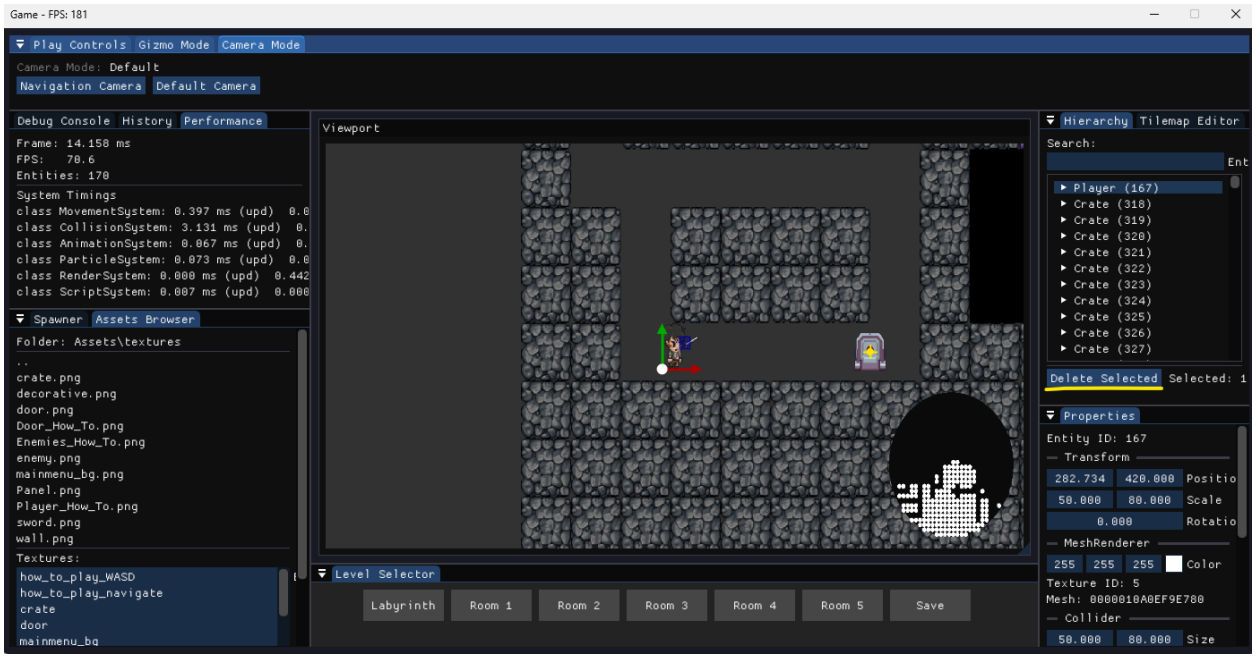
Usage:

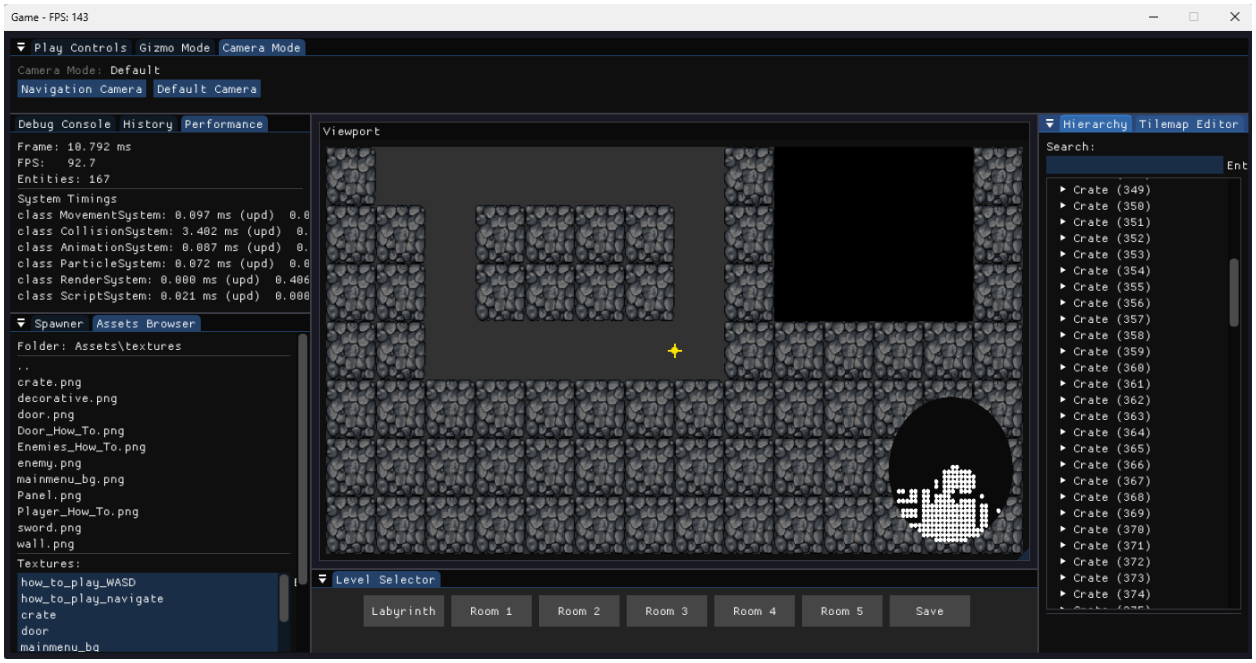
- Expand an entity node to see attached components (Transform, MeshRenderer, Collider, etc.).
- Click to select it for property editing.
- Press Delete Selected to remove.

Below we can see the properties of our created crate with the unique texture by clicking on it in the hierarchy panel.



And to delete entities, simply click on them and click delete selected.


Game - FPS: 181
Play Controls
Gizmo Mode
Camera Mode
Camera Mode: Default
Navigation Camera
Default Camera
Debug Console
History
Performance
Frame: 14.158 ms
FPS: 70.6
Entities: 170
System Timings
class MovementSystem: 0.397 ms (upd) 0.0
class CollisionSystem: 3.131 ms (upd) 0.
class AnimationSystem: 0.067 ms (upd) 0.
class ParticleSystem: 0.073 ms (upd) 0.0
class RenderSystem: 0.000 ms (upd) 0.442
class ScriptSystem: 0.007 ms (upd) 0.000
Spawner
Assets Browser
Folder: Assets\textures
..
crate.png
decorative.png
door.png
Door_How_To.png
Enemies_How_To.png
enemy.png
mainmenu_bg.png
Panel.png
Player_How_To.png
sword.png
wall.png
Textures:
how_to_play_WASD
how_to_play_navigate
crate
door
mainmenu_bg
Viewport
Level Selector
Labyrinth
Room 1
Room 2
Room 3
Room 4
Room 5
Save
Hierarchy
Tilemap Editor
Search:
Ent
Player (167)
Crate (318)
Crate (319)
Crate (320)
Crate (321)
Crate (322)
Crate (323)
Crate (324)
Crate (325)
Crate (326)
Crate (327)
Delete Selected
Selected: 1
Properties
Entity ID: 167
Transform
282.734
420.000
Positio
50.000
80.000
Scale
0.000
Rotatio
MeshRenderer
255
255
255
Color
Texture ID: 5
Mesh: 0000010A0EF9E780
Collider
50.000
80.000
Size


Game - FPS: 143
Play Controls
Gizmo Mode
Camera Mode
Camera Mode: Default
Navigation Camera
Default Camera
Debug Console
History
Performance
Frame: 10.792 ms
FPS: 92.7
Entities: 167
System Timings
class MovementSystem: 0.097 ms (upd) 0.0
class CollisionSystem: 3.402 ms (upd) 0.
class AnimationSystem: 0.087 ms (upd) 0.
class ParticleSystem: 0.072 ms (upd) 0.0
class RenderSystem: 0.000 ms (upd) 0.406
class ScriptSystem: 0.021 ms (upd) 0.000
Spawner
Assets Browser
Folder: Assets\textures
..
crate.png
decorative.png
door.png
Door_How_To.png
Enemies_How_To.png
enemy.png
mainmenu_bg.png
Panel.png
Player_How_To.png
sword.png
wall.png
Textures:
how_to_play_WASD
how_to_play_navigate
crate
door
mainmenu_bg
Viewport
Level Selector
Labyrinth
Room 1
Room 2
Room 3
Room 4
Room 5
Save
Hierarchy
Tilemap Editor
Search:
Ent
Crate (349)
Crate (350)
Crate (351)
Crate (352)
Crate (353)
Crate (354)
Crate (355)
Crate (356)
Crate (357)
Crate (358)
Crate (359)
Crate (360)
Crate (361)
Crate (362)
Crate (363)
Crate (364)
Crate (365)
Crate (366)
Crate (367)
Crate (368)
Crate (369)
Crate (370)
Crate (371)
Crate (372)
Crate (373)
Crate (374)

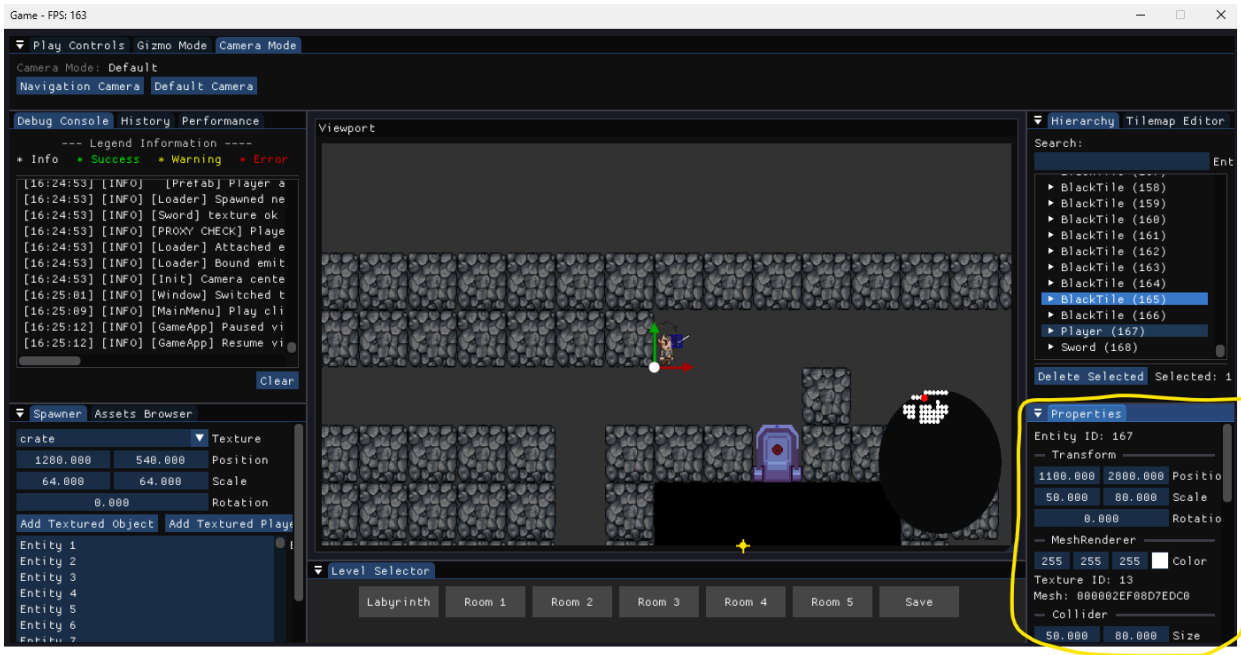
1.4 Property Editor

Appears under the Hierarchy panel.

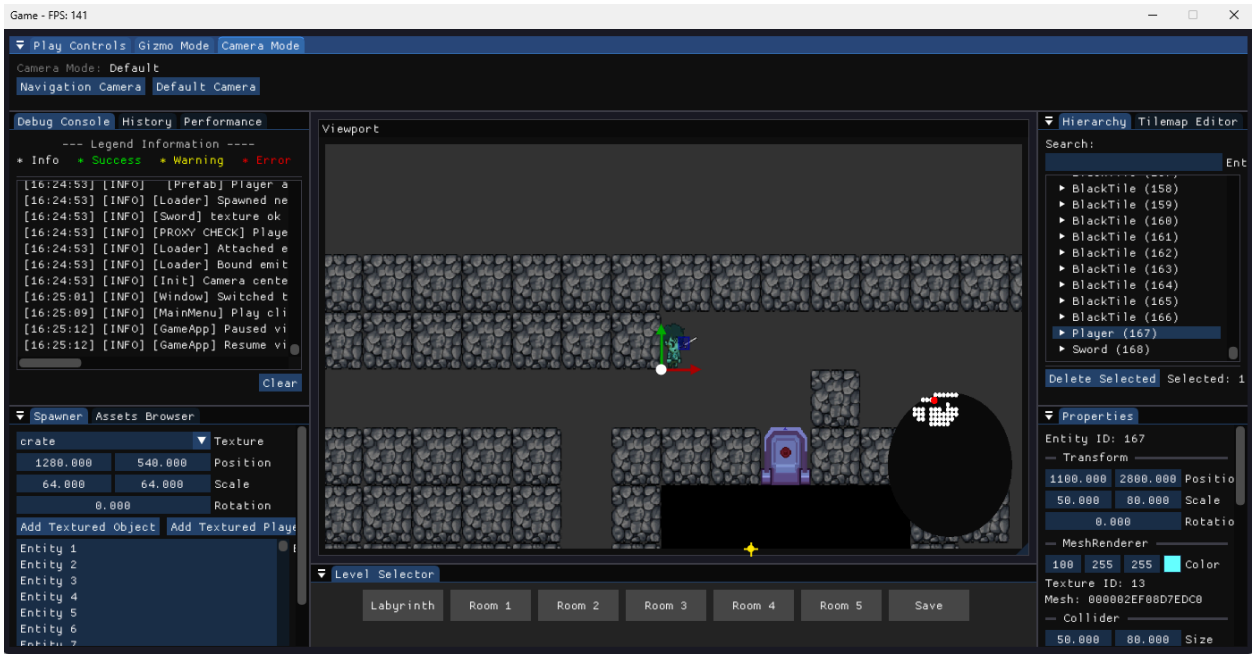
Purpose:
Allows fine-tuning of selected entity properties.

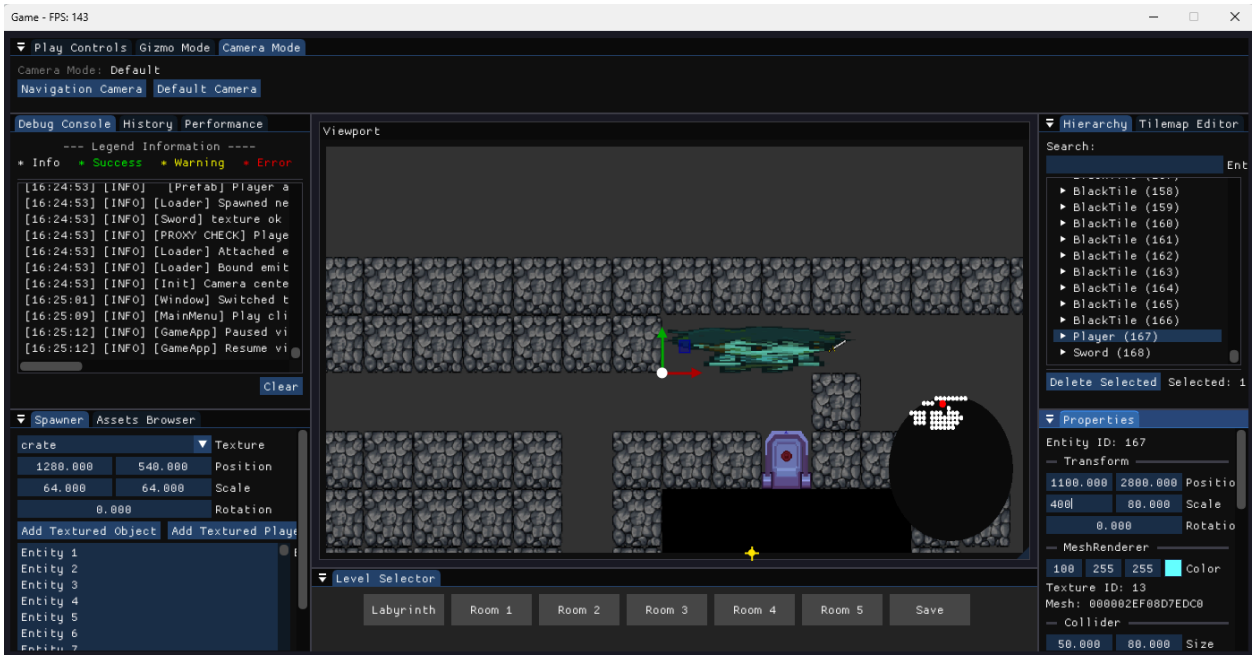
Editable Components:

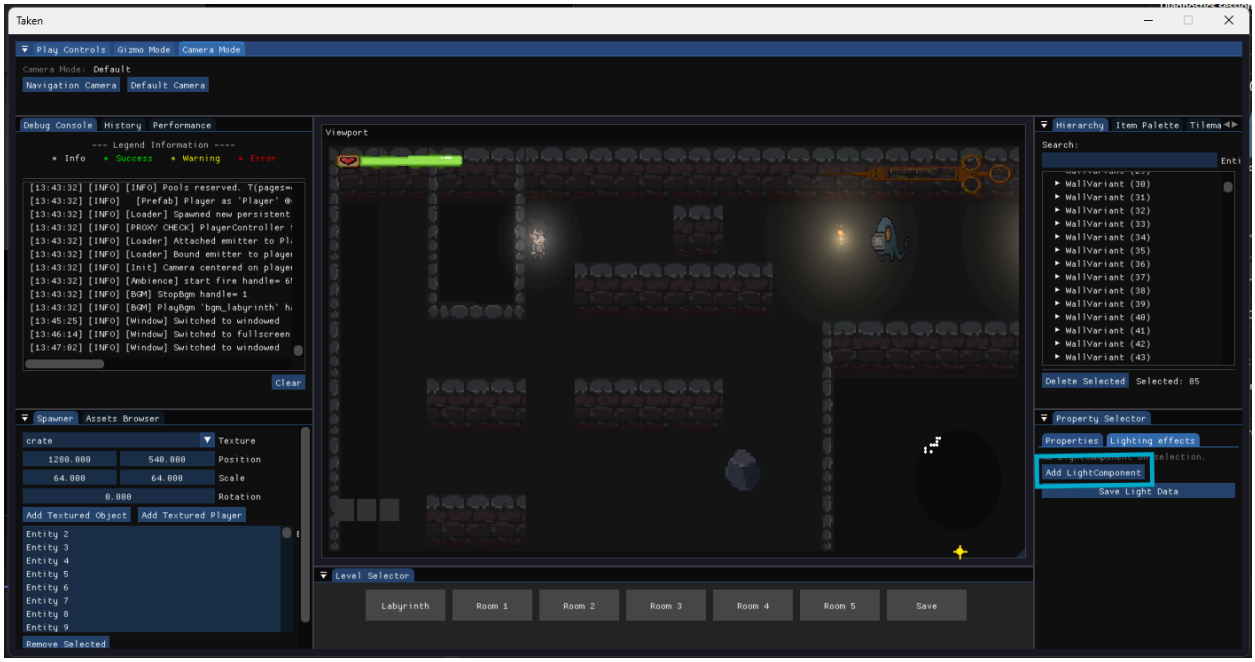
- Transform: Position, Scale, Rotation
- MeshRenderer: Color, Texture, Mesh reference
- Collider: Size, Trigger flag
- SpriteAnimator: Frames, Speed, Sheet info
- Lighting effects: Glow, light source, ember emitter, flicker



By editing the properties, we can directly change the properties of the entity's components.


Game - FPS: 141
Play Controls
Gizmo Mode
Camera Mode
Camera Mode: Default
Navigation Camera
Default Camera
Debug Console
History
Performance
--- Legend Information ----
Info
Success
Warning
Error
[16:24:53] [INFO] [Prefab] Player a
[16:24:53] [INFO] [Loader] Spawned ne
[16:24:53] [INFO] [Sword] texture ok
[16:24:53] [INFO] [PROXY CHECK] Playe
[16:24:53] [INFO] [Loader] Attached e
[16:24:53] [INFO] [Loader] Bound emit
[16:24:53] [INFO] [Init] Camera cente
[16:25:01] [INFO] [Window] Switched t
[16:25:09] [INFO] [MainMenu] Play cli
[16:25:12] [INFO] [GameApp] Paused vi
[16:25:12] [INFO] [GameApp] Resume vi
Clear
Spawner
Assets Browser
crate
Texture
1280.000
540.000
Position
64.000
64.000
Scale
0.000
Rotation
Add Textured Object
Add Textured Playe
Entity 1
Entity 2
Entity 3
Entity 4
Entity 5
Entity 6
Viewport
Level Selector
Labyrinth
Room 1
Room 2
Room 3
Room 4
Room 5
Save
Hierarchy
Tilemap Editor
Search:
BlackTile (158)
BlackTile (159)
BlackTile (160)
BlackTile (161)
BlackTile (162)
BlackTile (163)
BlackTile (164)
BlackTile (165)
BlackTile (166)
Player (167)
Sword (168)
Delete Selected
Selected: 1
Properties
Entity ID: 167
Transform
1100.000
2800.000
Position
50.000
80.000
Scale
0.000
Rotation
MeshRenderer
100
255
255
Color
Texture ID: 13
Mesh: 000002EF08D7EDC0
Collider
50.000
80.000
Size

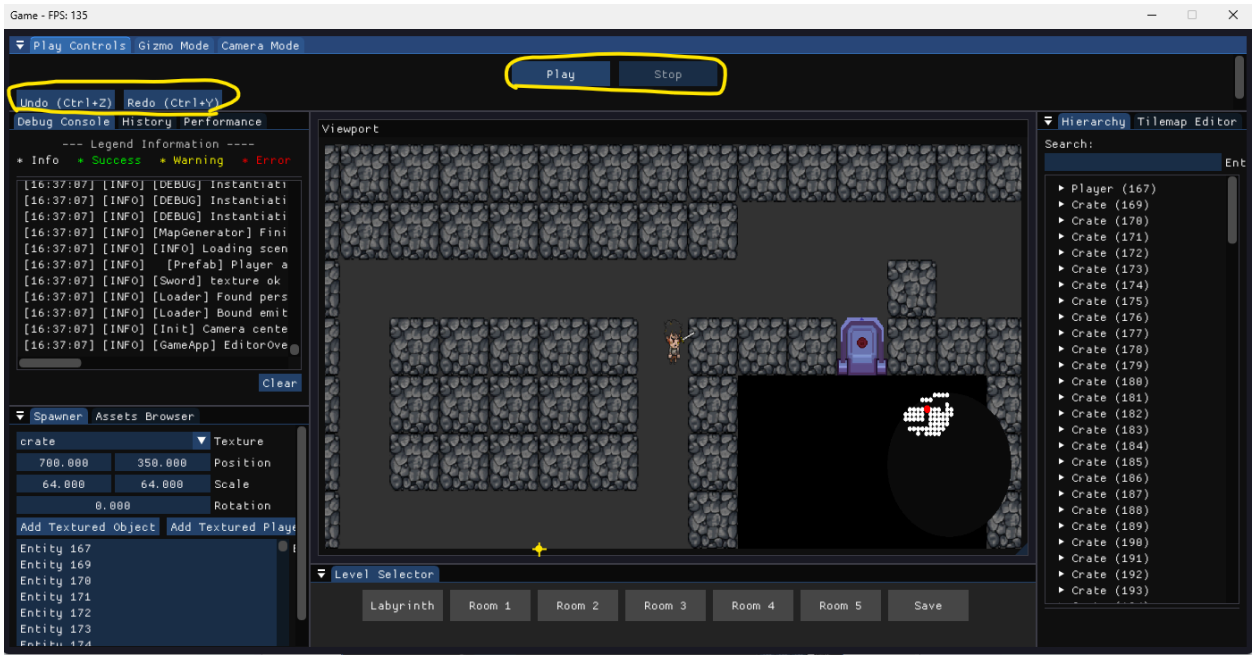

Game - FPS: 143
Play Controls
Gizmo Mode
Camera Mode
Camera Mode: Default
Navigation Camera
Default Camera
Debug Console
History
Performance
--- Legend Information ----
Info
Success
Warning
Error
[16:24:53] [INFO] [Prefab] Player a
[16:24:53] [INFO] [Loader] Spawned ne
[16:24:53] [INFO] [Sword] texture ok
[16:24:53] [INFO] [PROXY CHECK] Playe
[16:24:53] [INFO] [Loader] Attached e
[16:24:53] [INFO] [Loader] Bound emit
[16:24:53] [INFO] [Init] Camera cente
[16:25:01] [INFO] [Window] Switched t
[16:25:09] [INFO] [MainMenu] Play cli
[16:25:12] [INFO] [GameApp] Paused vi
[16:25:12] [INFO] [GameApp] Resume vi
Clear
Spawner
Assets Browser
crate
Texture
1280.000
540.000
Position
64.000
64.000
Scale
0.000
Rotation
Add Textured Object
Add Textured Playe
Entity 1
Entity 2
Entity 3
Entity 4
Entity 5
Entity 6
Viewport
Level Selector
Labyrinth
Room 1
Room 2
Room 3
Room 4
Room 5
Save
Hierarchy
Tilemap Editor
Search:
BlackTile (158)
BlackTile (159)
BlackTile (160)
BlackTile (161)
BlackTile (162)
BlackTile (163)
BlackTile (164)
BlackTile (165)
BlackTile (166)
Player (167)
Sword (168)
Delete Selected
Selected: 1
Properties
Entity ID: 167
Transform
1100.000
2800.000
Position
400
80.000
Scale
0.000
Rotation
MeshRenderer
100
255
255
Color
Texture ID: 13
Mesh: 000002EF08D7EDC0
Collider
50.000
80.000
Size

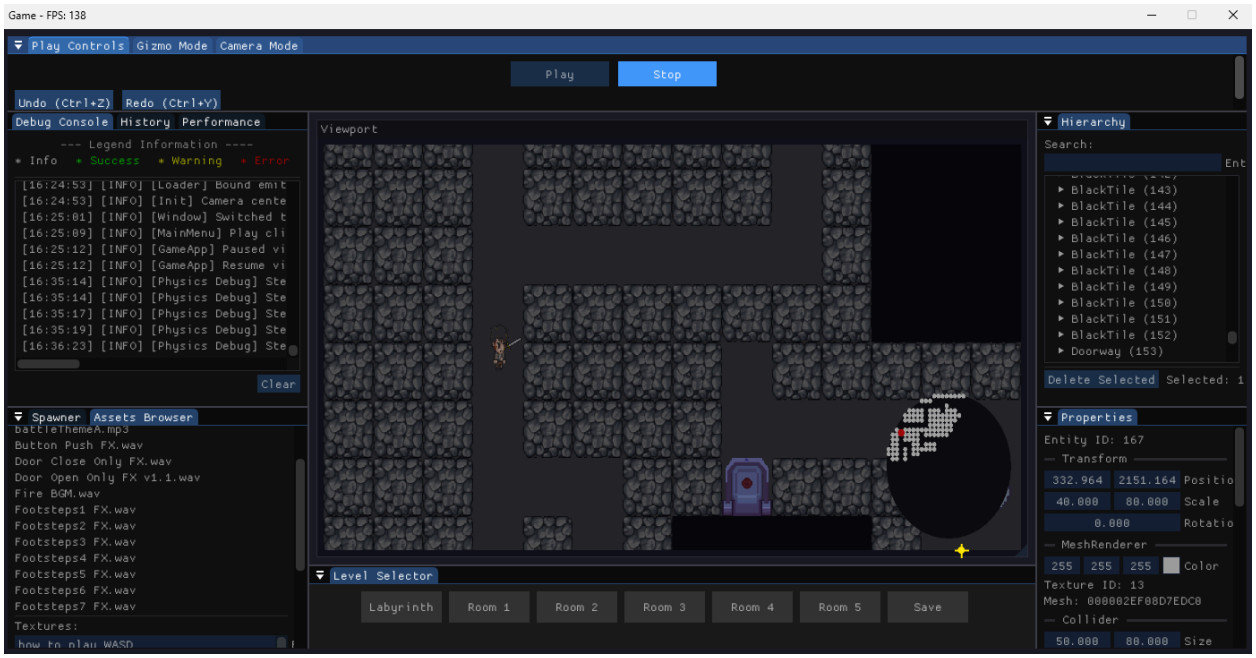

Taken
Play Controls
Gizmo Mode
Camera Mode
Camera Mode: Default
Navigation Camera
Default Camera
Debug Console
History
Performance
--- Legend Information ---
Info
Success
Warning
Error
[13:43:32] [INFO] [INFO] Pools reserved. T(pages=
[13:43:32] [INFO] [Prefab] Player as 'Player' @
[13:43:32] [INFO] [Loader] Spawned new persistent
[13:43:32] [INFO] [PROXY CHECK] PlayerController
[13:43:32] [INFO] [Loader] Attached emitter to Pl
[13:43:32] [INFO] [Loader] Bound emitter to playe
[13:43:32] [INFO] [Init] Camera centered on playe
[13:43:32] [INFO] [Ambience] start fire handle= 6
[13:43:32] [INFO] [BGM] StopBgm handle= 1
[13:43:32] [INFO] [BGM] PlayBgm 'bgm_labyrinth' h
[13:45:25] [INFO] [Window] Switched to windowed
[13:46:14] [INFO] [Window] Switched to fullscreen
[13:47:02] [INFO] [Window] Switched to windowed
Clear
Spawner
Assets Browser
crate
Texture
1280.000
540.000
Position
64.000
64.000
Scale
0.000
Rotation
Add Textured Object
Add Textured Player
Entity 2
Entity 3
Entity 4
Entity 5
Entity 6
Entity 7
Entity 8
Entity 9
Remove Selected
Viewport
Level Selector
Labyrinth
Room 1
Room 2
Room 3
Room 4
Room 5
Save
Hierarchy
Item Palette
Tilema
Search:
WallVariant (30)
WallVariant (31)
WallVariant (32)
WallVariant (33)
WallVariant (34)
WallVariant (35)
WallVariant (36)
WallVariant (37)
WallVariant (38)
WallVariant (39)
WallVariant (40)
WallVariant (41)
WallVariant (42)
WallVariant (43)
Delete Selected
Selected: 85
Property Selector
Properties
Lighting effects
Add LightComponent
Save Light Data

1.5 Play / Stop Bar

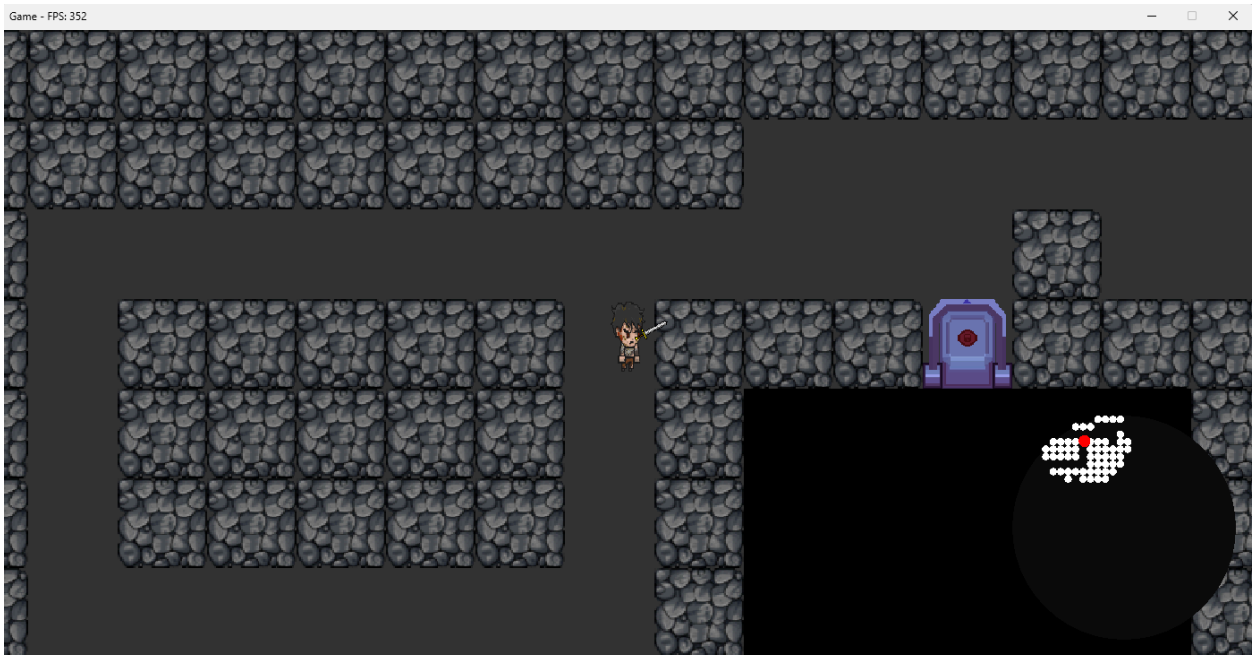
Top center bar for runtime simulation control.

- **Play**: switches the game into Play mode (locks editor updates), starts simulation.
- **Stop**: returns to editing mode.
- **Undo :** undo any edit done in steps
- **Redo :** undo any undos





Stop simply resets the player back to pre simulation position.

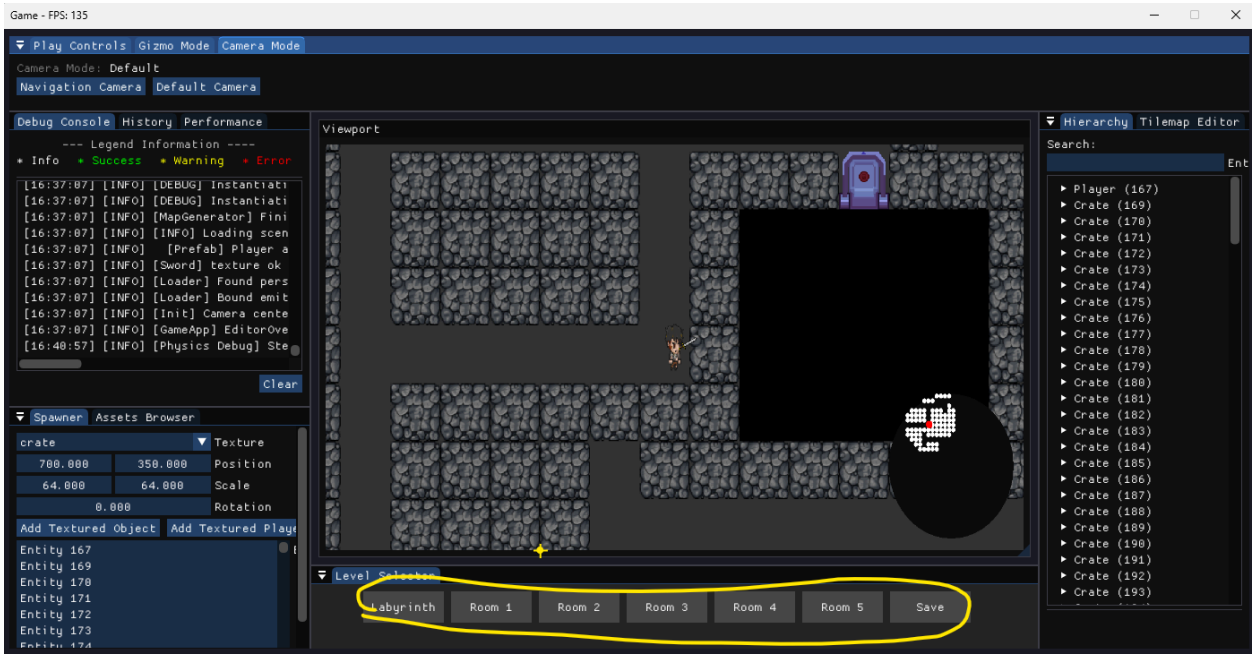


1.6 Level Selector

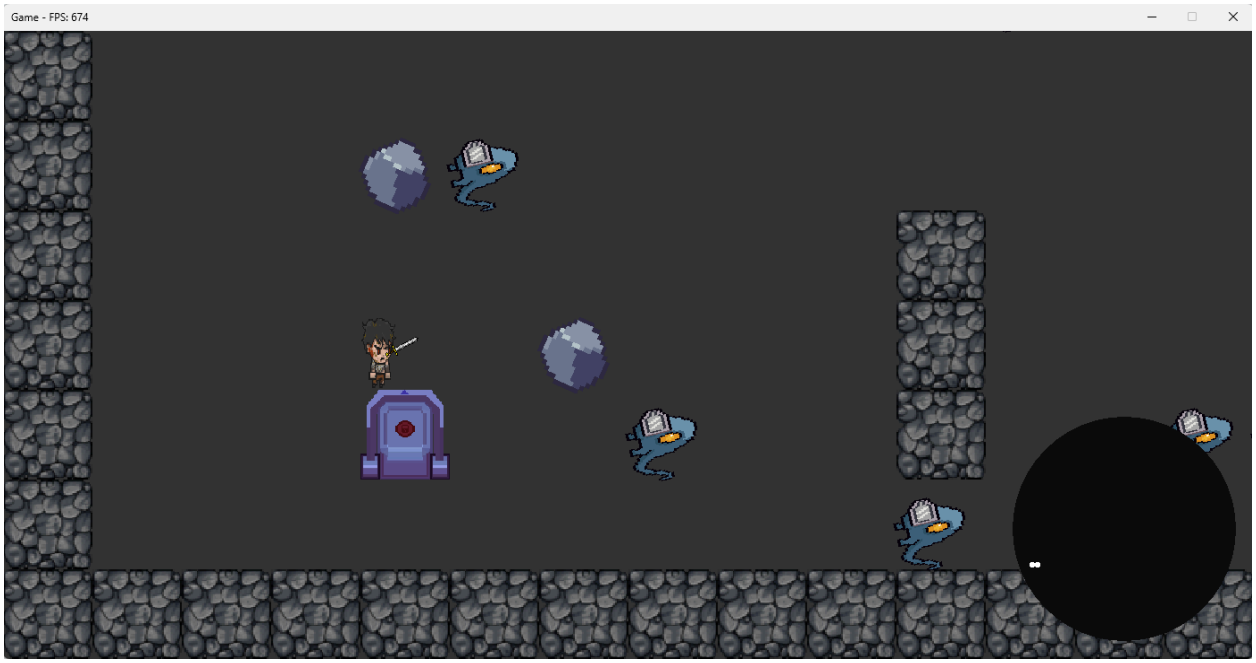
Bottom center of the viewport.

Usage:

- Buttons for each level (Labyrinth, Room 1–5).
- **Save** button writes all entities to the current level JSON.



Switches Levels Accordingly



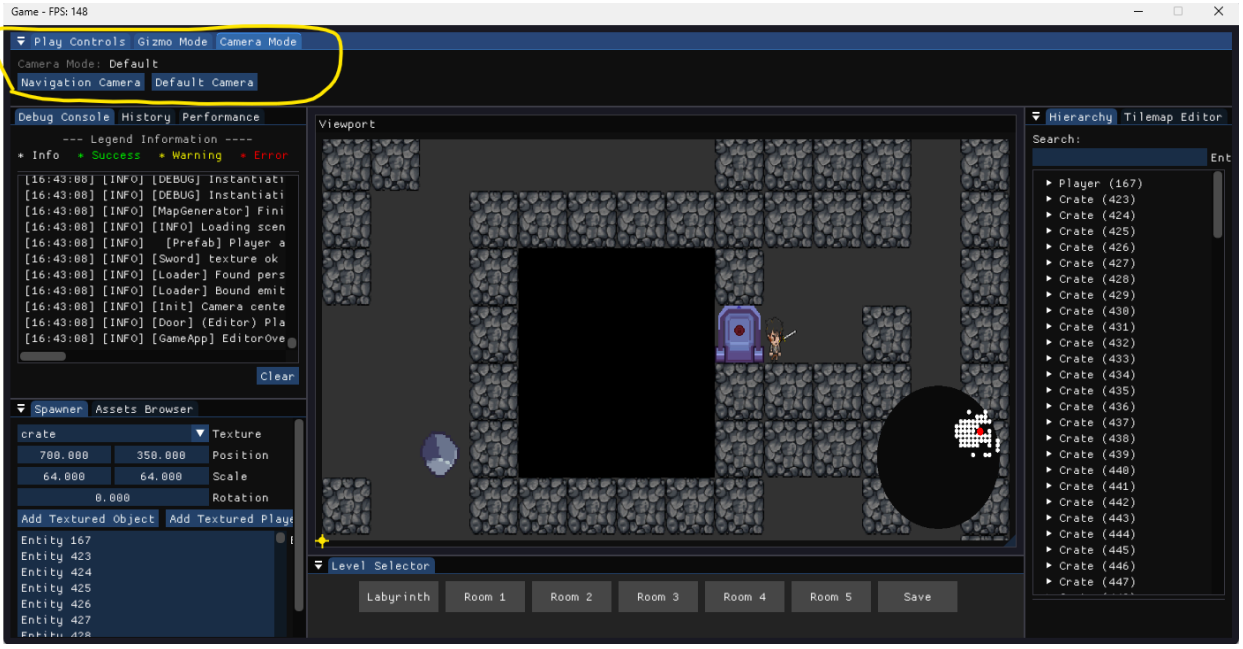
1.7 Camera Mode Panel

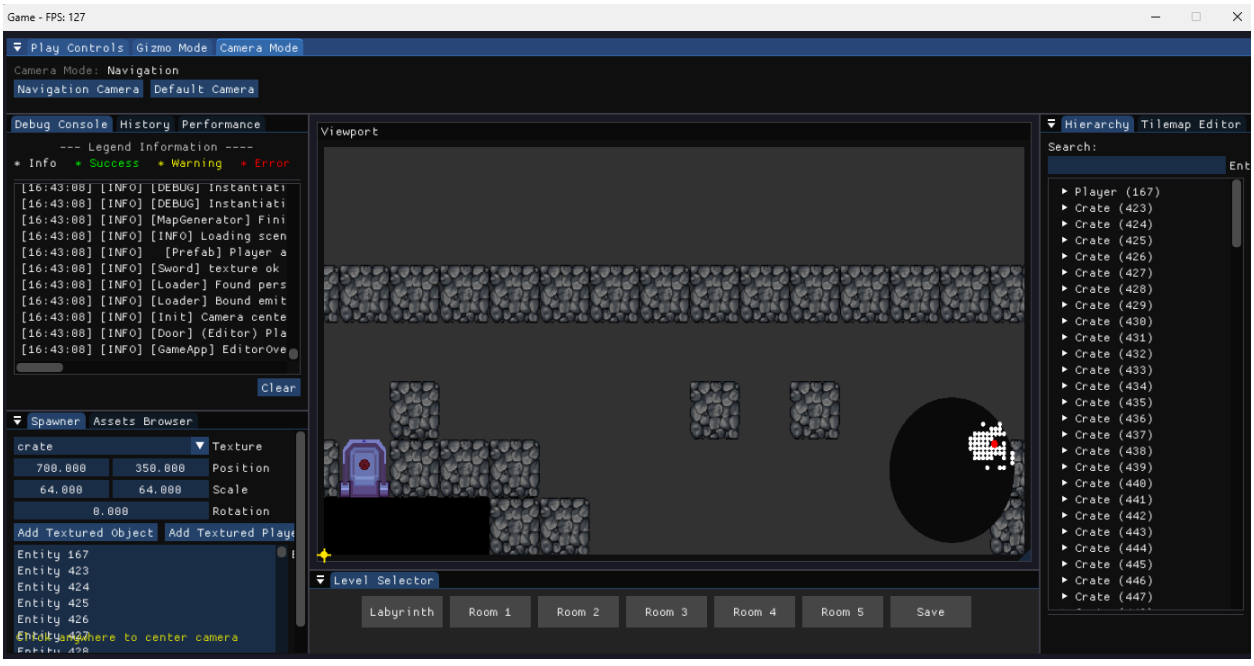
Bottom left corner.

Switch between:

- **Default Camera:** game camera follow mode.
- **Navigation Camera:** free-move editor camera (click anywhere to center).

A small tooltip appears when navigation mode is active.





1.8 Debug console

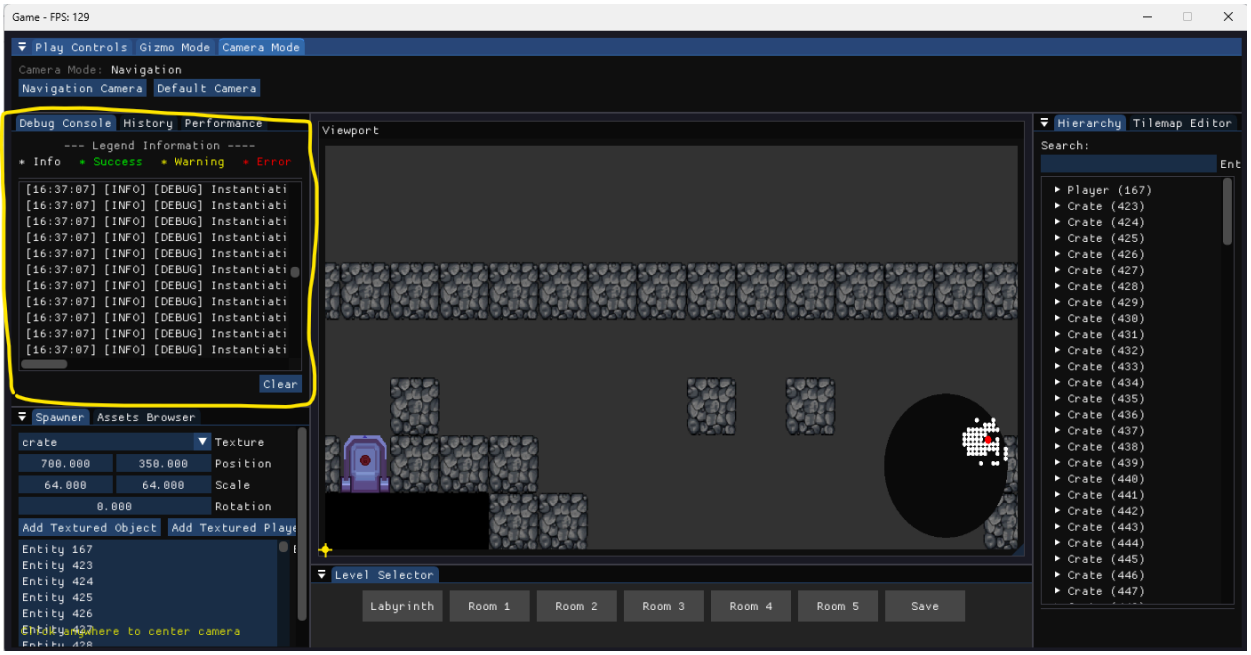
Purpose:
The debug console allows u to read all information without polluting the terminal

Description:
At the top left panel of the IMGUi, there is a debug panel, white text represents basic information, green text represents success stats, yellow text represents warnings and red text represents error .

There's also a clear button to clean all information, for a specific debug statement

This allows users to:

- **Debug** based on the information provided
- **Convenient** as there's no need to swap between game and terminal for debug purpose



1.9 Tile Map editor

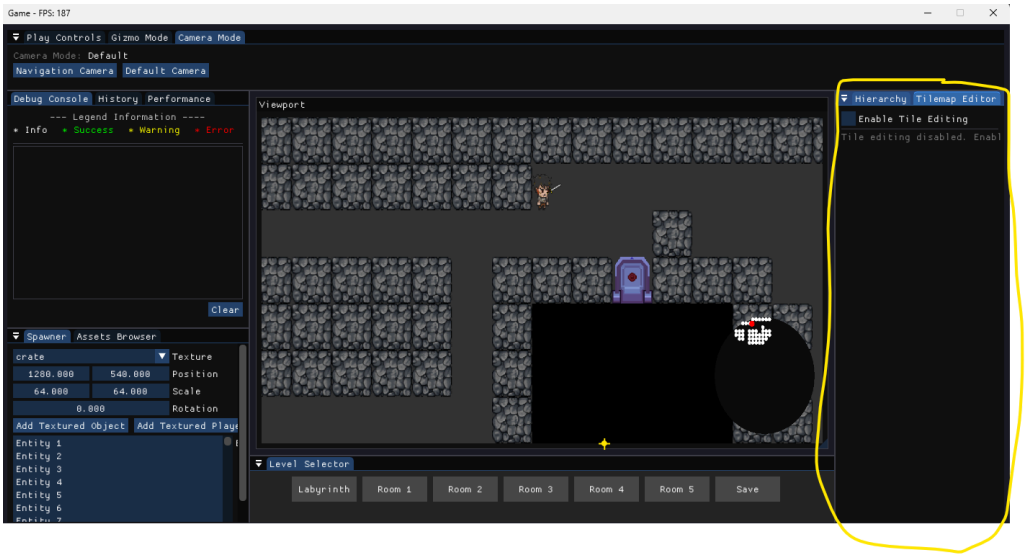
Purpose:

The Tilemap Editor allows users to visually create, modify, and manage 2D tile-based levels directly inside the engine, without manually editing any JSON or asset files. It gives a real-time, grid-based environment for drawing walls, crates, and empty spaces.

Description:

On the **right panel** of the IMGUI layout, there is a dedicated **Tilemap Editor tab**.
When **Enable Tile Editing** is checked, users can interact with the tile grid using the mouse.

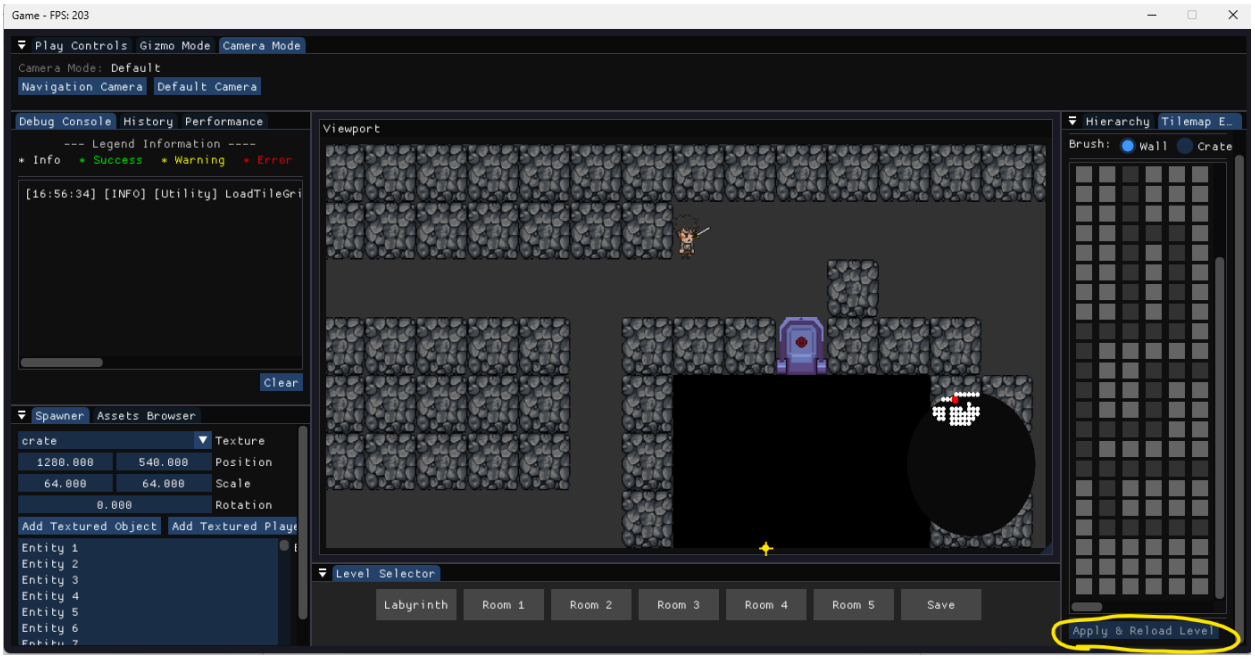
- The editor displays the **entire tile grid** as a matrix of squares.
- Each tile's color represents its current type:
    - **Dark tiles (black)** = blocked/solid areas
    - **Grey tiles** = walkable / empty
    - **Yellow tiles** = crates or interactables
- A **Brush selection toolbar** is provided at the top:
    - **Wall** – paints solid wall tiles
    - **Crate** – places crate-type tiles (yellow)
    - **Empty** – clears a tile, turning it into walkable space
- Clicking on the tile grid will immediately update the map shown in the central **Viewport**, providing real-time visual feedback.
- The scrollbar on the right allows navigating large maps



Pressing enable tile map



Press apply and reload level to save changes



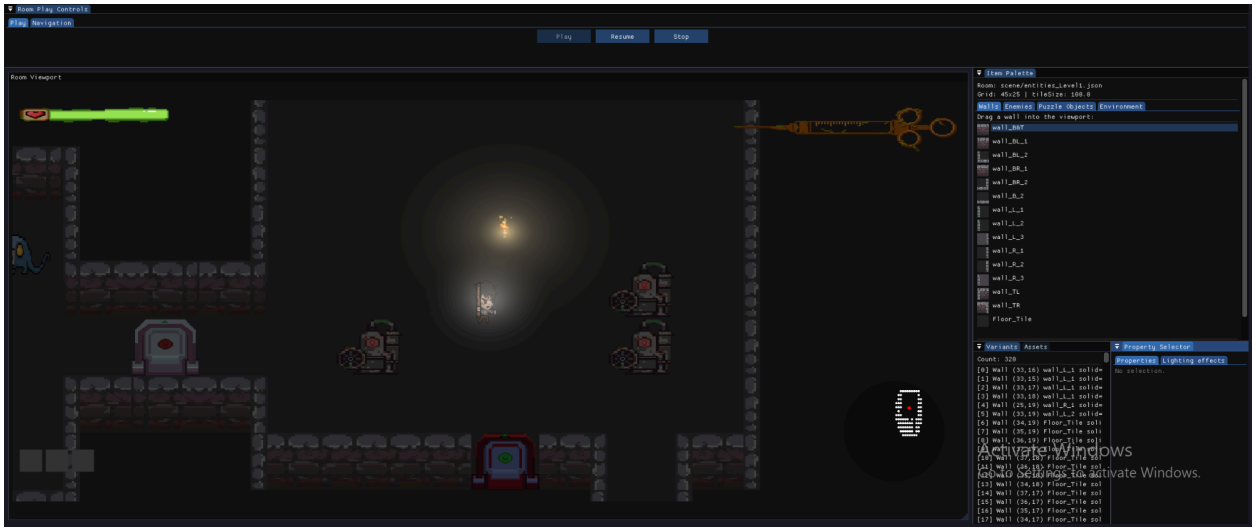
========== **RoomEditor** ==========
==========

It allows users to:

- Navigate around the room with the camera system
- Delete and adding assets from file explorer
- Drag and drop walls with different textures
- Switch between *Editor* and *Play* modes
- Edit room at runtime
- Multi Select with bulk delete
- Paintbrush tool
- Property selector

Interface Layout

Below is the layout of our RoomEditor, toggled on and off via the f1 button, it is also auto docked

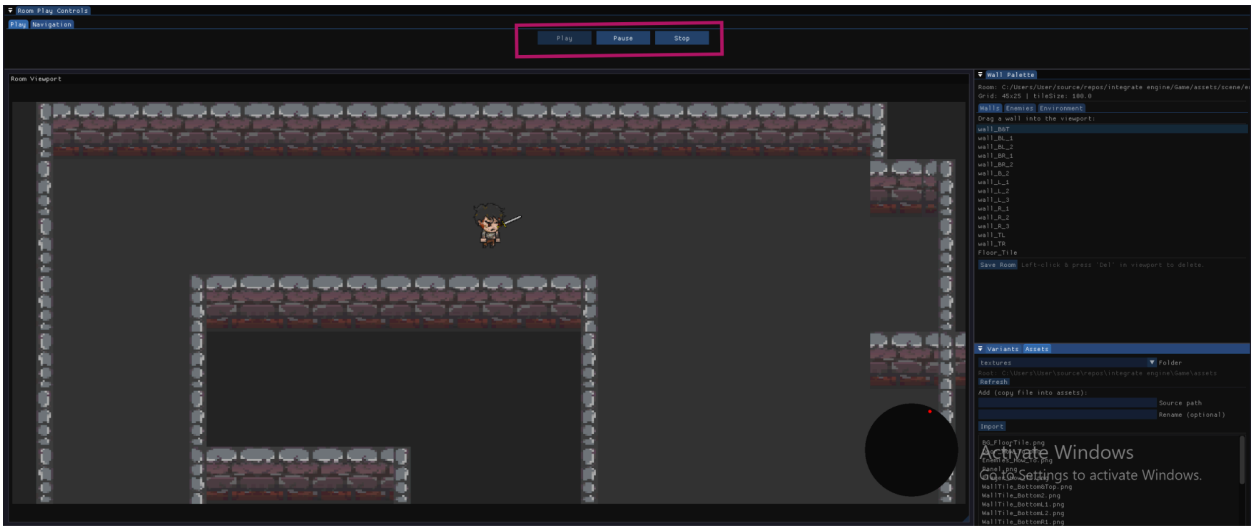


Panel Descriptions

2.0 Play / Pause/ Stop Bar

Top center bar for runtime simulation control.

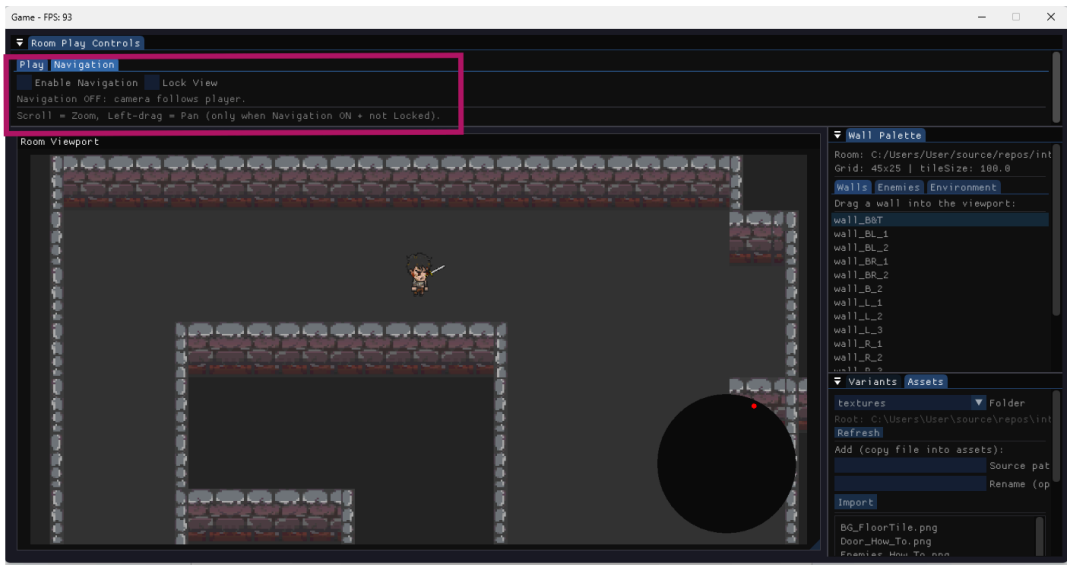
- **Play**: switches the game into Play mode (locks editor updates), starts simulation.
- **Pause**: returns to editing mode.
- **Stop**: returns to original position in the main map



2.1 Navigation system

Top left bar for camera navigation

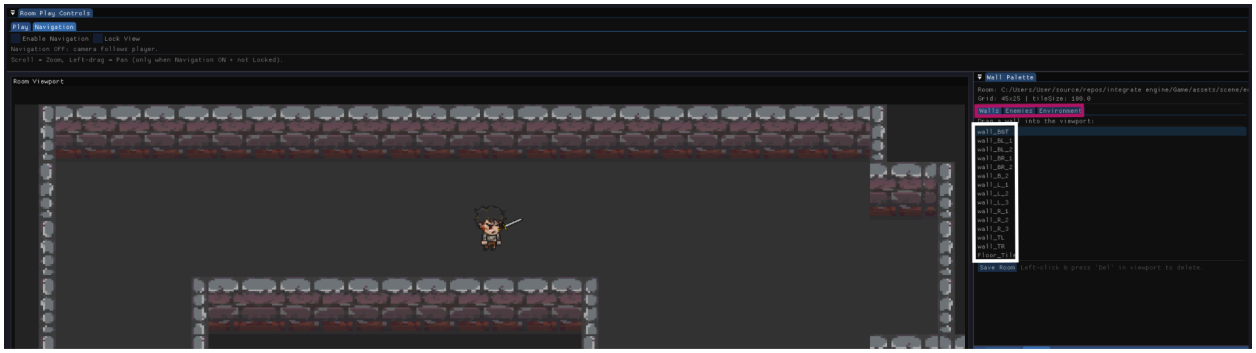
- Enable navigation ✅ = 1) left click drag to roam the rooms
- 2) scroll wheel to zoom/un -zoom
- Lock view ✅ = Freezes the current screen + allow other interaction of Imgui tools



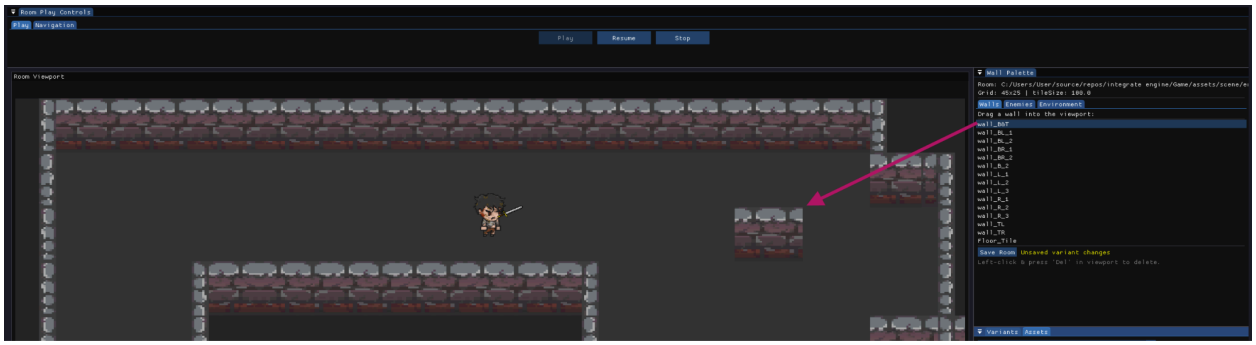
2.2 Drag and Drop of items

Top right bar for room designs at runtime

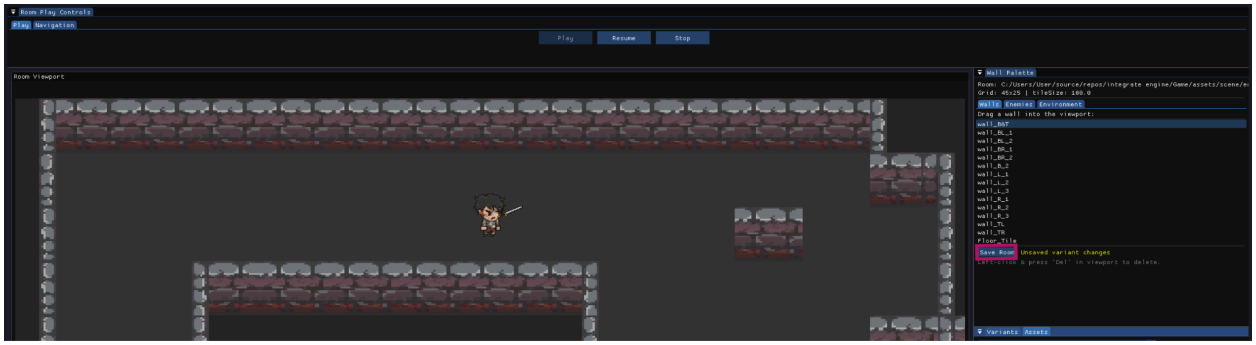
- Select appropriate items according to designs (walls, Enemies, Environment)
- Drag it out into the viewport
- Press save to save the designs
- View all the different variants in the room
- To delete click on item and press del



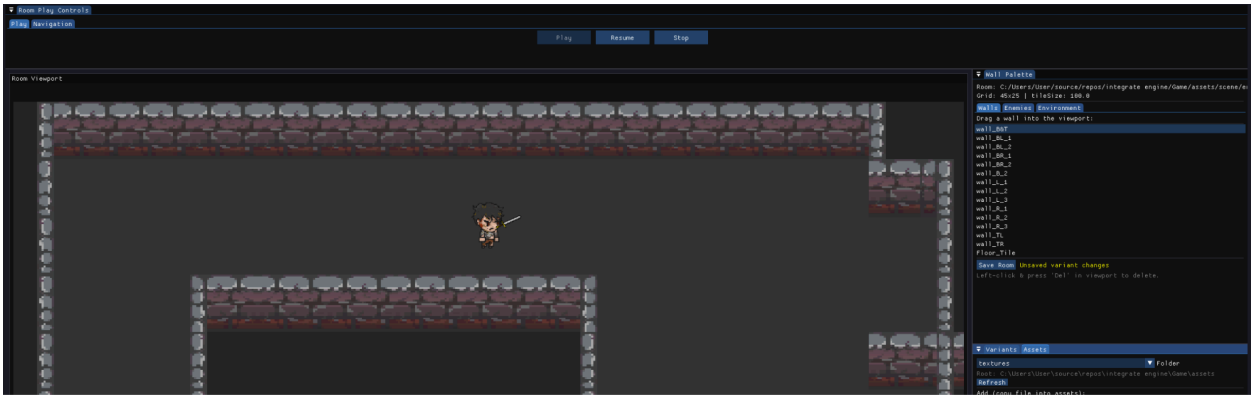
Drag and drop into viewport



Save it



Delete it through left click and del

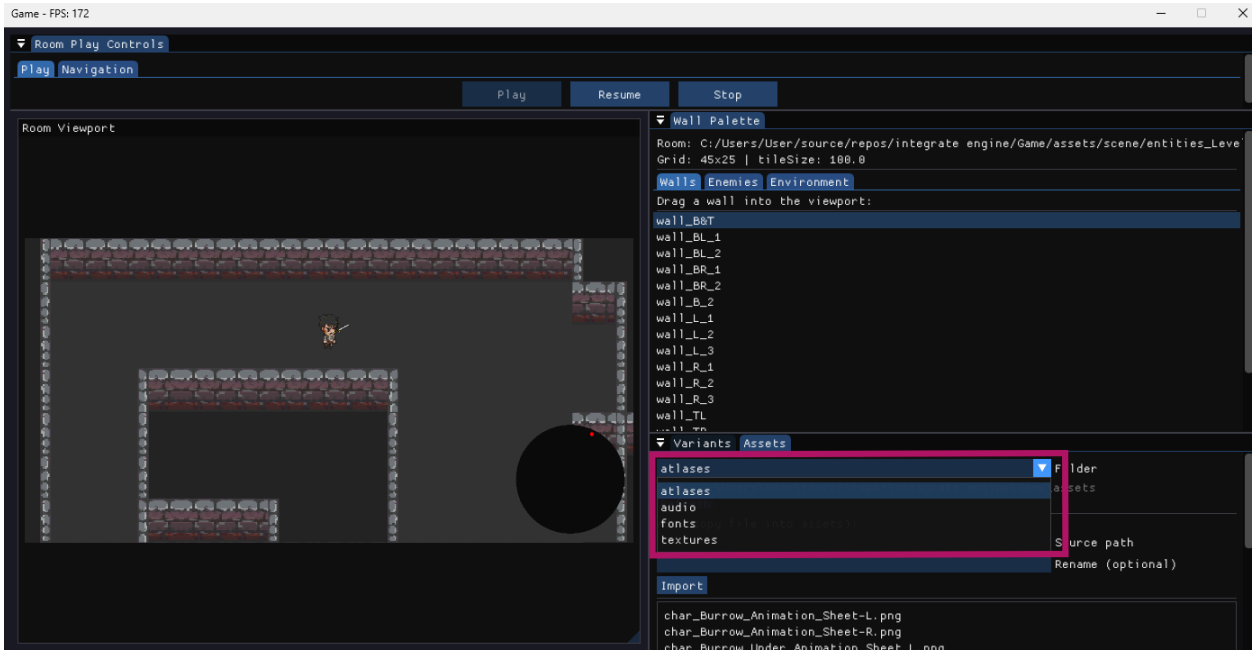


2.3 Deleting/adding of new assets

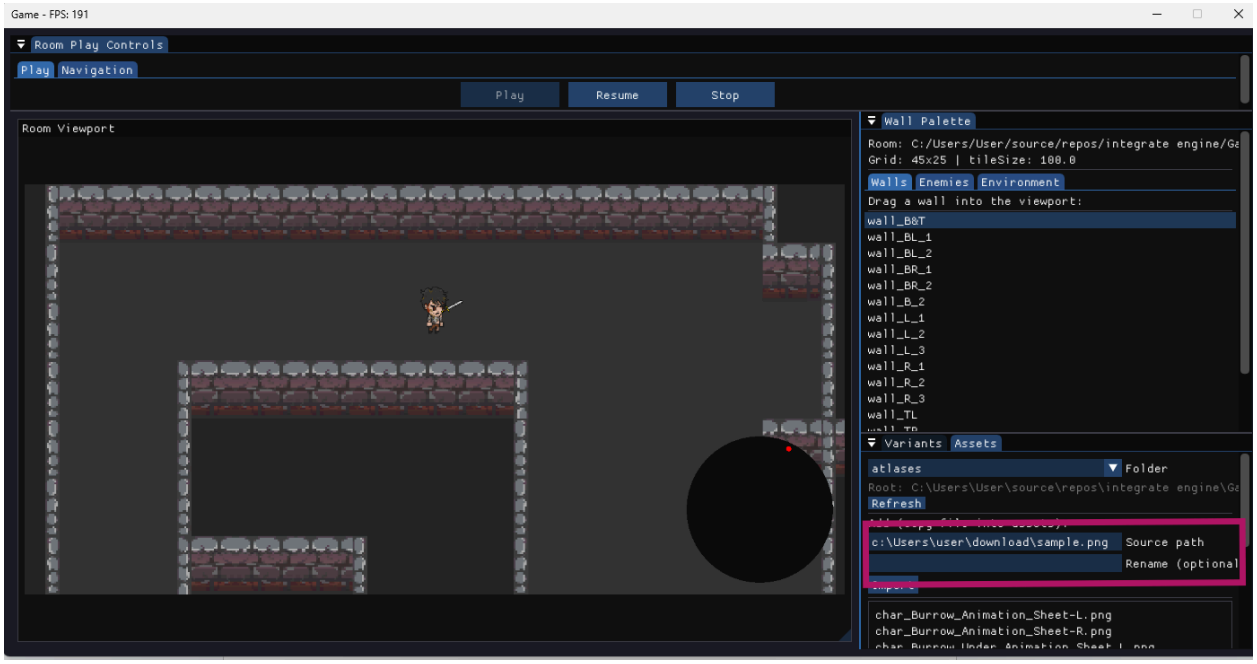
Bottom right panel for adding/deleting assets in the file explorer

- Click on what type of assets to import/delete
- Each folder contain its own type of assets (textures, spritesheet, audio, font)
- To add simply type in the full path of the assets into the prompt
- Changing name is optional
- To delete, find the assets click on it and scroll to the bottom and press delete

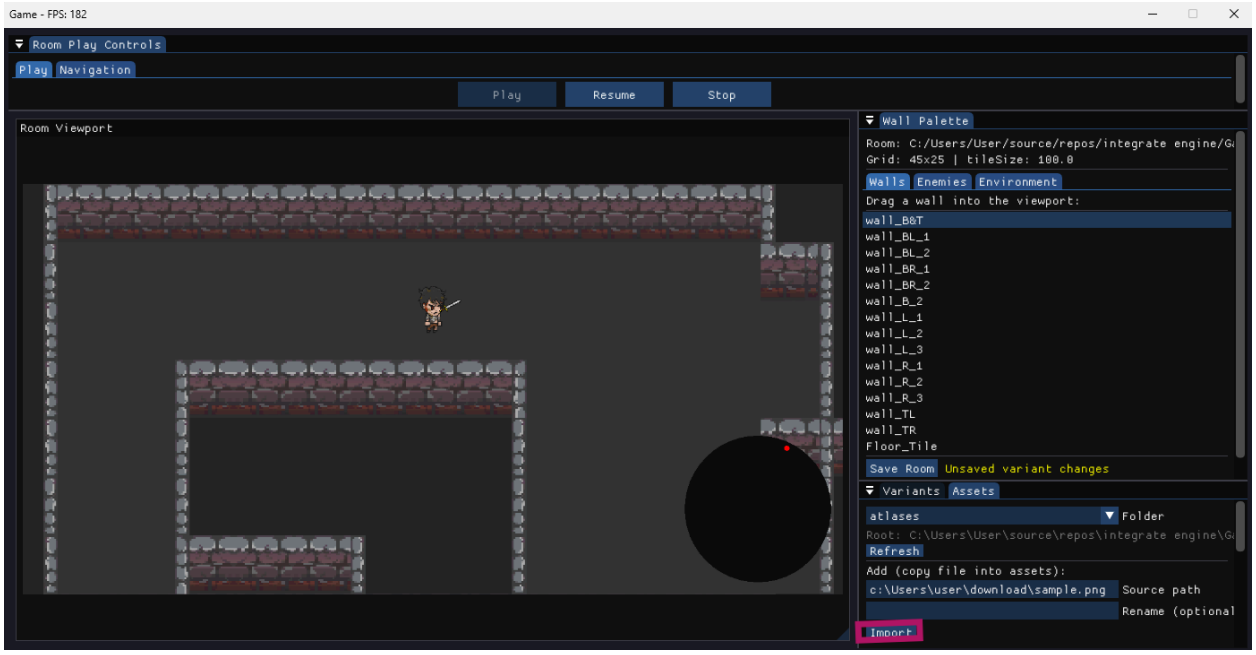
Find the correct assets type



Type in full of assets according to the location in the hardware

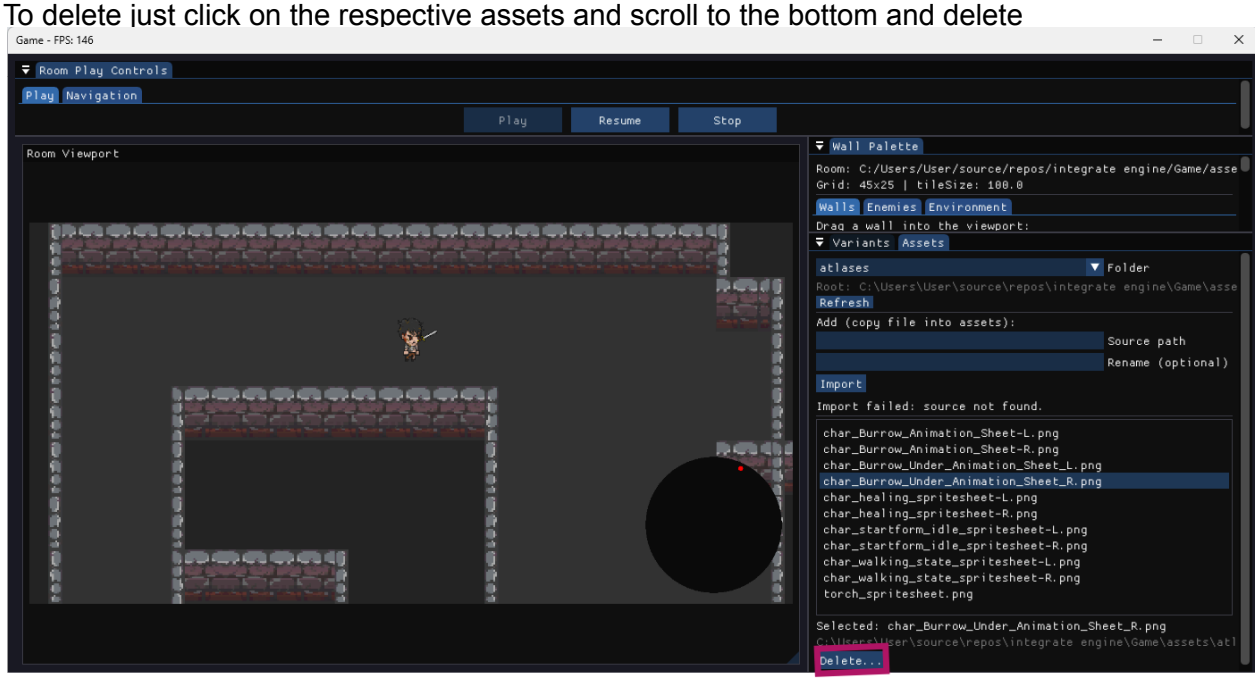


Press import



If path doesnt match, “fail to import” will appear
If there is such path, it will be reflected onto the editor

To delete just click on the respective assets and scroll to the bottom and delete



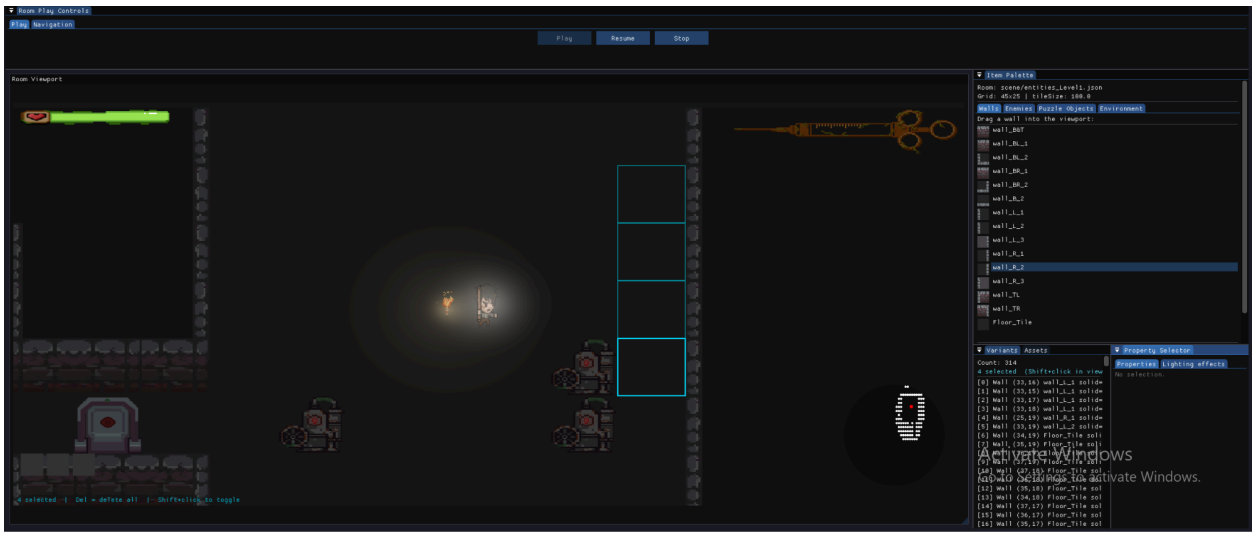
There will be an confirmation to delete the assets or not

2.4 Multi Select with bulk delete

Supports Shift+click to select multiple grid cells simultaneously. Selected cells are highlighted with cyan outlines in the viewport and highlighted rows in the variant list panel. Pressing Delete removes all selected entities at once.



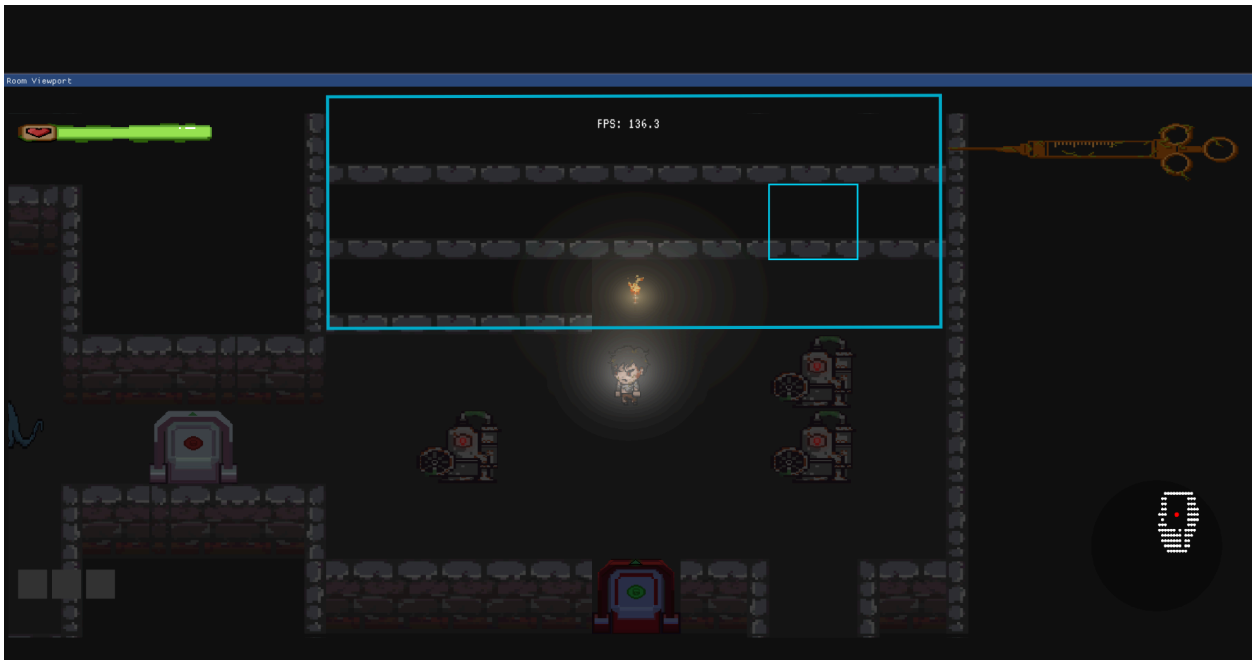
Shift click + multiple entities



Pressing delete

2.5 Paintbrush tool

A paintbrush mode (toggled with [P]) allows the editor to paint wall textures or prefabs continuously by holding the left mouse button and dragging across the viewport, instead of dragging and dropping each tile individually. The currently armed item is selected by clicking any entry in the Walls, Enemies, or Environment palette tabs. A green cell highlight follows the mouse as a cursor preview, and door cells are protected with an orange highlight. The painted result is immediately reflected in both the live ECS and the grid data.

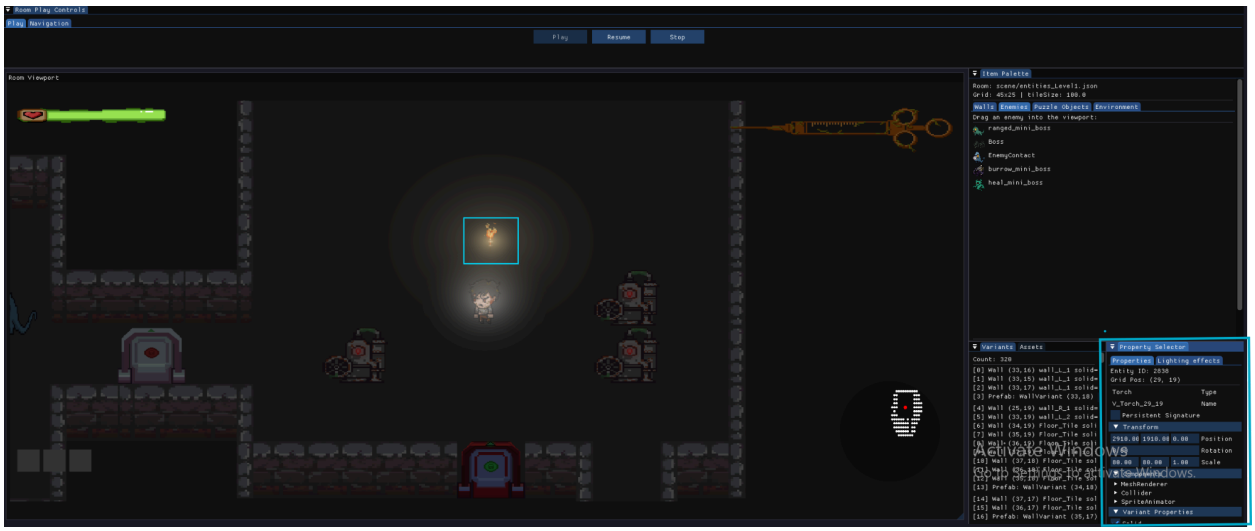


2.5 Property selector

Appears at the bottom right of the roomEditor

Components:

- Transform: Position, Scale, Rotation
- MeshRenderer: Color, Texture, Mesh reference
- Collider: Size, Trigger flag
- SpriteAnimator: Frames, Speed, Sheet info
- Lighting effects: Glow, light source, ember emitter, flicker



Click on any entities and start modifying